Formulario e costanti

Coulomb	$F = k\,\dfrac{\lvert q_1 q_2\rvert}{r^2} \qquad k = \dfrac{1}{4\pi\varepsilon_0}$
Campo elettrico	$E = \dfrac{F}{q_0} \qquad F = qE \qquad E = k\,\dfrac{\lvert Q\rvert}{r^2}$ (carica puntiforme)
Flusso	$\Delta\Phi = E\,\Delta S\cos\theta \qquad \Phi_\Omega = \displaystyle\sum_i \vec{E}_i \cdot \Delta\vec{S}_i$
Teorema di Gauss	$\Phi_\Omega = \dfrac{Q_{\text{int}}}{\varepsilon_0}$
Filo infinito	$E = \dfrac{\lambda}{2\pi\varepsilon_0\, d} = \dfrac{2k\lambda}{d}$
Piano infinito	$E = \dfrac{\sigma}{2\varepsilon_0}$
Sfera uniforme	$E = \dfrac{kQ\,r}{R^3}$ $(r < R) \qquad E = \dfrac{kQ}{r^2}$ $(r > R)$
Geometria	$S_{\text{sfera}} = 4\pi r^2 \qquad V_{\text{sfera}} = \frac{4}{3}\pi r^3 \qquad S_{\text{cilindro, laterale}} = 2\pi r h$
Dinamica	$\rho = \dfrac{m}{V} \qquad P = mg \qquad F = ma$

$k = 8{,}99 \times 10^9\,\text{N}\,\text{m}^2/\text{C}^2$ $\qquad$ $\varepsilon_0 = 8{,}85 \times 10^{-12}\,\text{C}^2/(\text{N}\,\text{m}^2)$

$e = 1{,}602 \times 10^{-19}\,\text{C}$ $\qquad$ $g = 9{,}8\,\text{m/s}^2$

Prefissi: $1\,\text{nC} = 10^{-9}\,\text{C}$, $1\,\mu\text{C} = 10^{-6}\,\text{C}$, $1\,\mu\text{m} = 10^{-6}\,\text{m}$, $1\,\text{cm} = 10^{-2}\,\text{m}$.